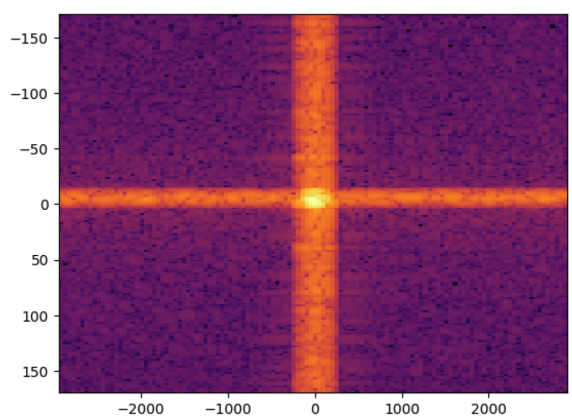
4️⃣

Systematics as a multiplicative component

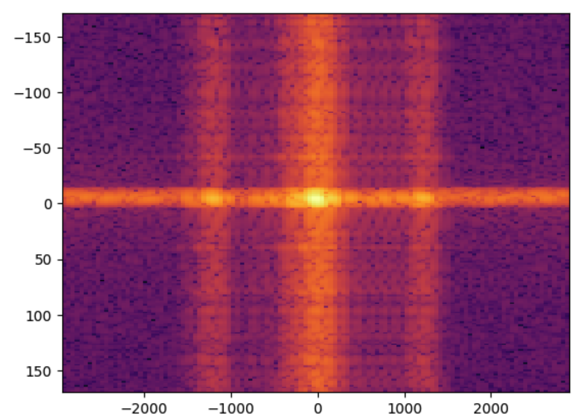
▼ Motivation:

What went wrong:

We found out that the systematics component really didn't look like an additive component the way we thought. Instead it looks like a multiplicative feature with much wider affects across the data space. An example is below:



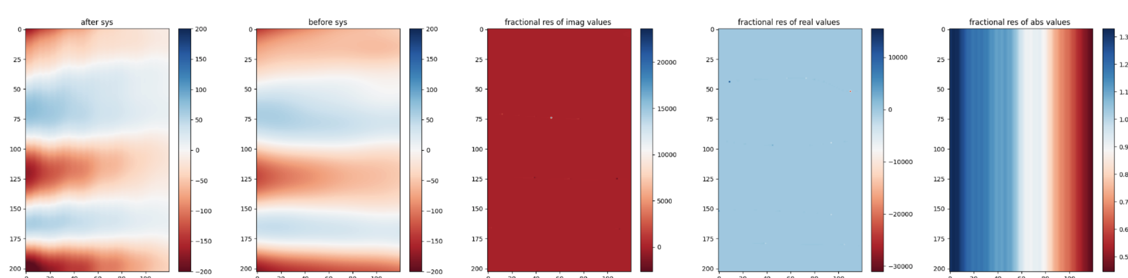
Original data without systematics

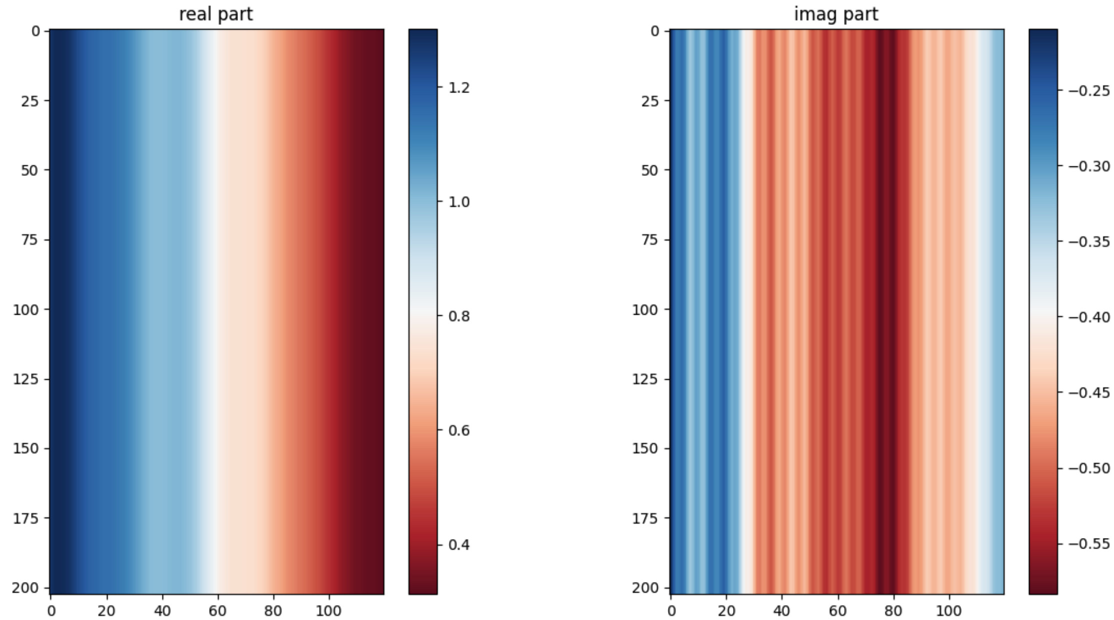


Data with systematic effects

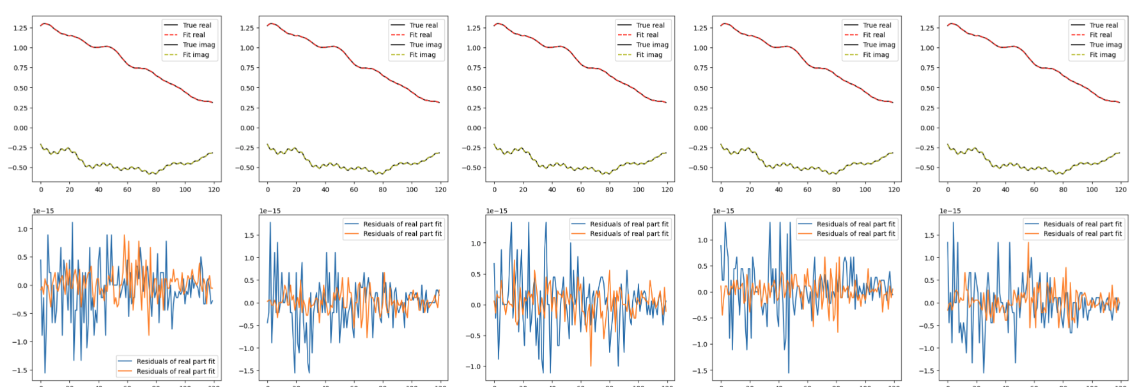
▼ Tests:

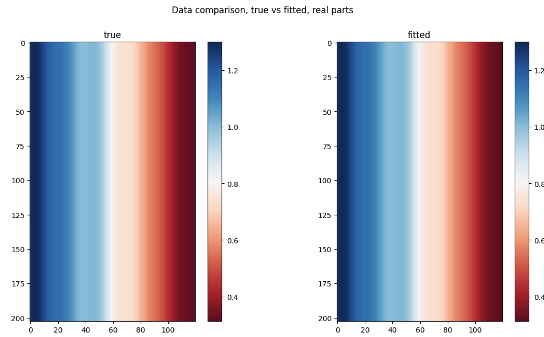
We looked at the plots of the ratio of $data_{frac} = \frac{\text{visibilities-with-systematics}}{\text{visibilities-without-systematics}}$ and got the following plot. We also looked at how the data at each FR varies with delay (aka frequency) and got the adjacent plot. This variation remains consistent across FR.

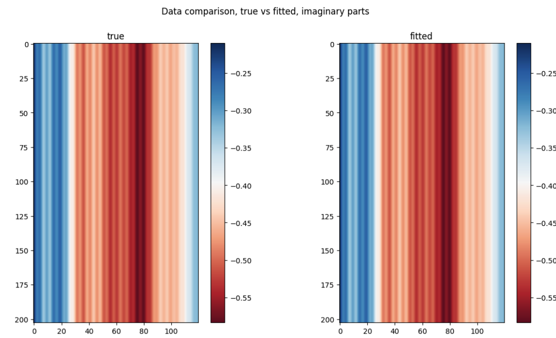


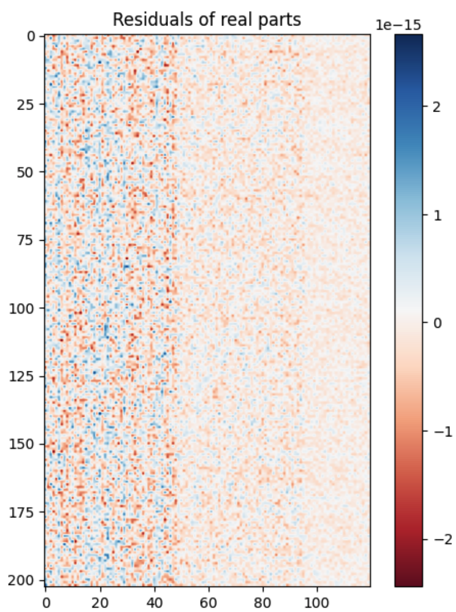


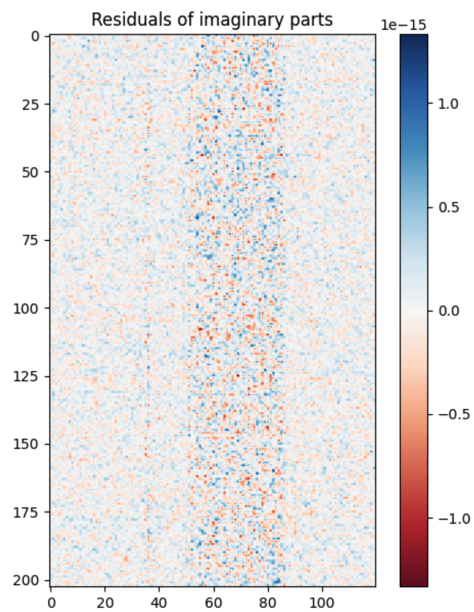
Now we tried to fit some coefficients using least squares such that $coeff\ @\ \mathbf{H} \propto data_{frac}$ to prove that we can indeed fit **H** to the data in some form. The results are below.

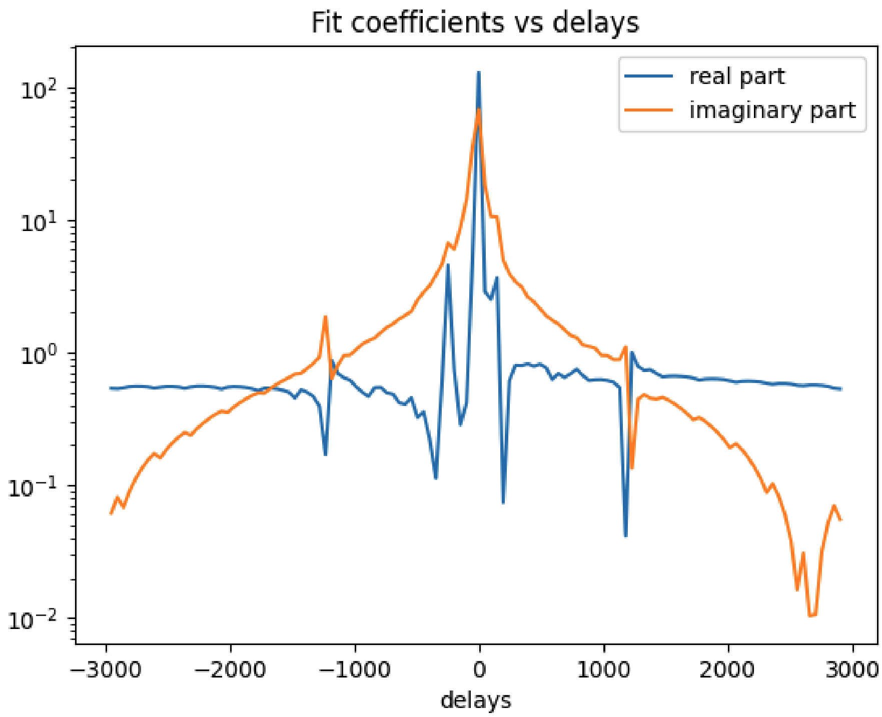



Data comparison, true vs fitted, real parts
true
fitted


Data comparison, true vs fitted, imaginary parts
true
fitted


Residuals of real parts
1e−15


Residuals of imaginary parts
1e−15



> ***Note:*** We are working with absolute values of the visibilities `np.abs(vis)` when creating $data_{frac}$.

▼ Reworking the Maths:

▼ The systematics component:

The starting data model now is: $d = y.(e + G.a_{fg}) + n$ where $y = H \,@\, b_{sys}$ and $e + G.a_{fg}$ will be written as s.

The systematics component of the equation that was earlier:

$$\left[\mathbf{B}^{-1} + \mathbf{H}^{\dagger}\mathbf{N}^{-1}\mathbf{H}\right]\mathbf{b}_{\mathbf{sys}} = \left[\mathbf{H}^{\dagger}\mathbf{N}^{-1}\mathbf{d} + \mathbf{H}^{\dagger}\mathbf{N}^{-1/2}\omega\right]$$

Now becomes the following, after we replace *H* by *s.H* from the data model above:

To implement this in the code, this had to be realified using the formalism defined in 0 Realification Maths. The equations implemented in the code are described below (LHS and RHS separately):

LHS:

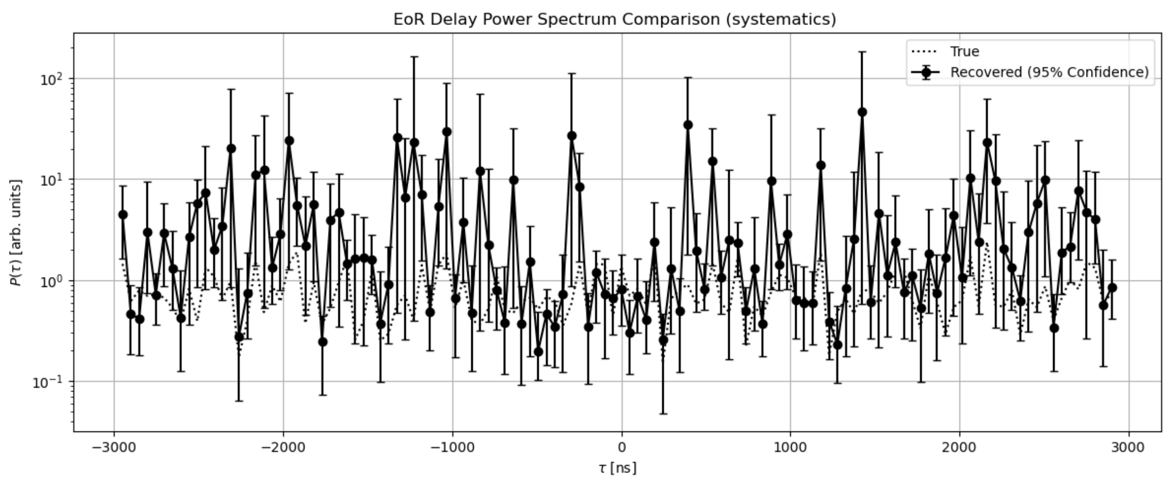
$$\left[\begin{bmatrix} B^{-1} & 0 \\ 0 & B^{-1} \end{bmatrix} + \begin{bmatrix} H_{re}^T & H_{im}^T \\ -H_{im}^T & H_{re}^T \end{bmatrix} \begin{bmatrix} s_{re}^T & s_{im}^T \end{bmatrix} \begin{bmatrix} N^{-1} & 0 \\ 0 & N^{-1} \end{bmatrix} \begin{bmatrix} s_{re} \\ s_{im} \end{bmatrix} \begin{bmatrix} H_{re} & -H_{im} \\ H_{im} & H_{re} \end{bmatrix}\right] \begin{bmatrix} b_{sys_{re}} \\ b_{sys_{im}} \end{bmatrix}$$

RHS:

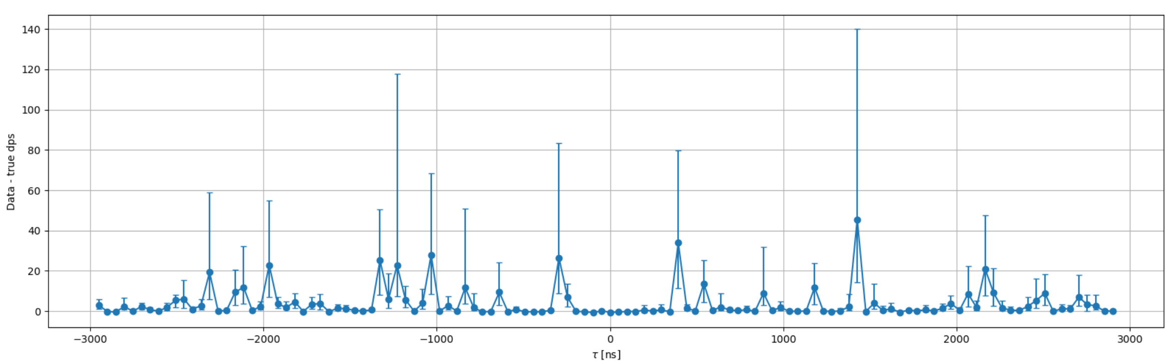
RHS had to be expanded and implemented in the following form due to computational challenges:

$$\begin{bmatrix} H_{re}^T s_{re}^T N^{-1} d_{re} + H_{im}^T s_{im}^T N^{-1} d_{im} + H_{re}^T s_{re}^T N^{-1/2} \omega_{re} + H_{im}^T s_{im}^T N^{-1/2} \omega_{im} \\ -H_{im}^T s_{re}^T N^{-1} d_{re} + H_{re}^T s_{im}^T N^{-1} d_{im} - H_{im}^T s_{re}^T N^{-1/2} \omega_{re} + H_{re}^T s_{im}^T N^{-1/2} \omega_{im} \end{bmatrix}$$

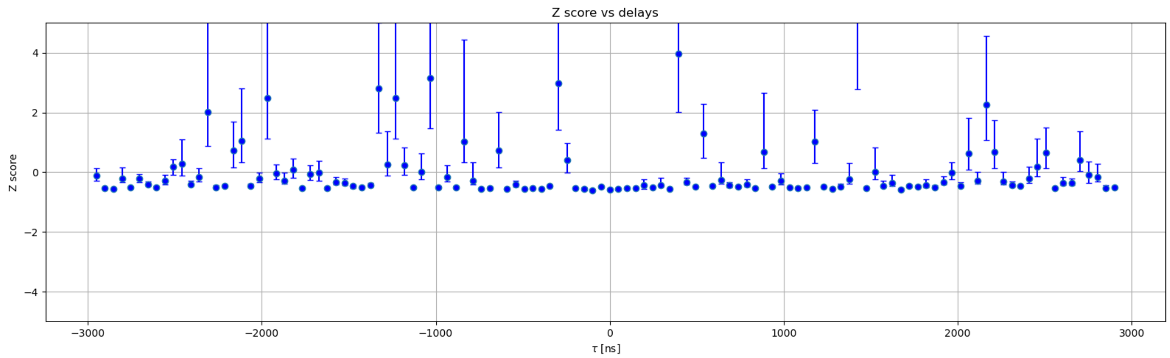
If we just include this update in the code, we get the following ***Results:***



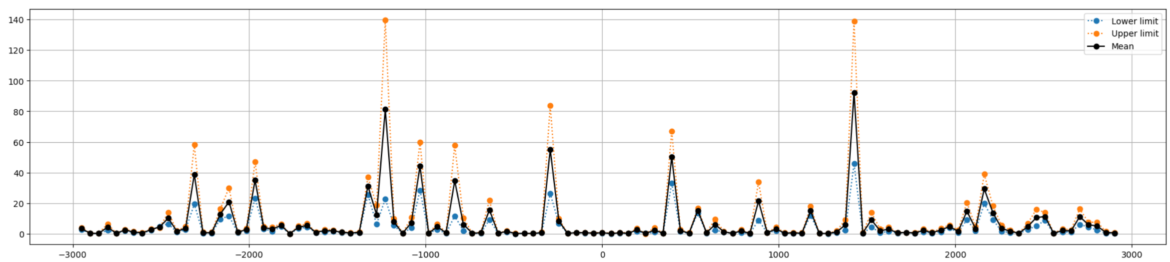
Delay power spectrum from this approach



Residuals against delays



Z scores against delays



Tracing the mean, lower and upper limits of the errorbars.

R2 score of the estimated DPS and true DPS: -369.38723118601604

▼ The EoR and FG components

The bad fit score (R2) inspired a revisit into the math. Due to the now dependent nature of the s and b_{sys} components in the data, we can no longer treat the three components as a joint linear system. So, we need to update the EoR-FG GCR as well to incorporate the effect of the systematics on the EoR-FG combined signal as well. This follows a similar format to what we found in the systematics case. Essentially, all the projection operators (including the identity operator for EoR) is multiplied by the systematic model defined by $y = H \,@\, b_{sys}$. The transformations are as follows:

EoR projection: $I —> yI$

FG projection: $G —> yG$

So the new GCR system of equations becomes:

$$\begin{bmatrix} E^{-1} + y^\dagger N^{-1} y & y^\dagger N^{-1} y G \\ G^\dagger y^\dagger N^{-1} y & G^\dagger y^\dagger N^{-1} y G \end{bmatrix} \begin{bmatrix} e \\ a_{fg} \end{bmatrix} = \begin{bmatrix} y^\dagger N^{-1} d + E^{-1/2}\omega_e + y^\dagger N^{-1/2}\omega_n \\ G^\dagger y^\dagger N^{-1} d + G^\dagger y^\dagger N^{-1/2}\omega_n \end{bmatrix}$$

Issues with this approach:

1. Haven't found an efficient way of implementing the above equation in the code yet (the only mathematically feasible way, where the shapes work, involves giant matrix multiplications that take forever)

2. The prior resampling issue: The inv_gamma distribution gets skewed, the rejection sampling never works because the samples drawn from the distribution never fall within the prior bounds.